The cubic Fejér property on simplices in every dimension

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Abstract

We prove the Fejér inequality for Bos's cubic simplex interpolation nodes in every dimension. The proof reduces the deficit to nonnegative coefficients after eliminating a least barycentric coordinate. An exact finite certificate and its complete coefficient table accompany the argument.

Scope and status. Only the cubic Fejér conjecture in AIM 641 is addressed. The three quartic conjectures grouped into that entry remain unaddressed. This is a solution claim submitted for review. Pre-submission mathematical, bibliographic, and editorial checks were performed with OpenAI Codex (AI); independent human review and formal verification are not claimed.

1 Problem and proof

We settle the cubic assertion within entry 641 [1]. Bos proved it through dimension 28 and described a dimension-by-dimension degree-elevation test [2]. The point here is to carry out the induction symbolically in the dimension, reducing all remaining dimensions to 45 exact coefficient types.

Let $S_d = \operatorname{conv}\{v_1, \dots, v_{d+1}\}$ be a nondegenerate real simplex. In barycentric coordinates its cubic node set is

$$F_{3,d} = \{e_i\}_i \ \cup \ \{te_i + (1-t)e_j : i \neq j\} \ \cup \ \{(e_i + e_j + e_k)/3 : i < j < k\}, \qquad t = \frac{1 + 1/\sqrt{5}}{2},$$

where e_i is the ith coordinate vector and an unavailable node type is omitted. For $M = d + 1$, the number of distinct nodes is

$$M + M(M-1) + \binom{M}{3} = \binom{d+3}{3} = \dim \mathbb{P}_3(\mathbb{R}^d).$$

Write $\ell_x^{(3,d)}$ for the cardinal polynomial at a node x. The formulas below also establish unisolvence directly.

Theorem 1.1. *For every nondegenerate real d-simplex, $d \geq 1$, Bos's cubic node set $F_{3,d}$ satisfies*

$$\sum_{x \in F_{3,d}} |\ell_x^{(3,d)}(z)|^2 \leq 1 \qquad (z \in S_d).$$

1.1 A dimension-independent symmetric deficit

Let $M = d+1$ be the number of barycentric coordinates. Work homogeneously with nonnegative $x_1, \dots, x_M$ and write

$$S = \sum_i x_i, \qquad p_j = \sum_i x_i^j.$$

The homogeneous cubic cardinal polynomials are

$$\ell_i = \frac{x_i}{2}(12x_i^2 - 12Sx_i + 3S^2 - p_2), \tag{1}$$

$$\ell_{ij} = 5x_ix_j\left(\frac{3+\sqrt{5}}{2}x_i + \frac{3-\sqrt{5}}{2}x_j - S\right), \quad i \neq j, \tag{2}$$

$$\ell_{ijk} = 27x_ix_jx_k, \quad i < j < k. \tag{3}$$

On $S = 1$, the indices i, ij, and ijk refer respectively to the three node types just defined. Direct substitution verifies the cardinal values: each polynomial equals one at its own node and zero at every other node. Here $t(1-t) = 1/5$, and the edge polynomial ℓ_{ij} is oriented toward $x_i = t$. Since the number of polynomials is the dimension of the polynomial space, these cardinal values also prove unisolvence. These are the homogenized polynomials from Bos's construction [2, Section 3]. Their squared sum, denoted $\mathcal{K}_M$, is

$$\begin{aligned}\mathcal{K}_M = {} & 54p_6 + 78Sp_5 + 29S^2p_4 - \frac{391}{2}p_2p_4 - 18S^3p_3 - 144Sp_2p_3 + 50p_3^2 \\ & + \frac{9}{4}S^4p_2 + \frac{47}{2}S^2p_2^2 + \frac{487}{4}p_2^3.\end{aligned} \tag{4}$$

For example, the face contribution follows from

$$729\sum_{i<j<k} x_i^2x_j^2x_k^2 = \frac{243}{2}p_2^3 - \frac{729}{2}p_2p_4 + 243p_6.$$

The vertex and edge contributions are obtained by the same direct expansion. Formula (4) is also checked by a direct exact expansion in six variables. This suffices in all dimensions: both expressions are symmetric homogeneous polynomials of degree six, are unchanged on appending a zero coordinate, and each monomial involves at most six distinct variables. Thus every possible coefficient is represented in that check.

Define the homogeneous deficit

$$H_M(x) = S^6 - \mathcal{K}_M(x).$$

The desired inequality is $H_M \geq 0$ on the nonnegative orthant, followed by restriction to $S = 1$. Appending a zero coordinate leaves the formula unchanged:

$$H_M(x_1, \dots, x_{M-1}, 0) = H_{M-1}(x_1, \dots, x_{M-1}).$$

1.2 One induction step covers the whole orthant

By symmetry, permute any nonnegative x so that $x_M = \min_i x_i$. Set

$$z = Mx_M, \qquad \mu_i = x_i - x_M \geq 0 \quad (1 \leq i < M).$$

Then

$$x_i = \mu_i + z/M \ (i < M), \qquad x_M = z/M. \tag{5}$$

Thus the centroid-to-facet substitution does not omit any point of the orthant.

Put $q_j = \sum_{i=1}^{M-1} \mu_i^j$. Under (5),

$$p_j = \sum_{i=1}^{j} \binom{j}{i} q_i (z/M)^{j-i} + \frac{z^j}{M^{j-1}}, \qquad 1 \le j \le 6, \tag{6}$$

so $S = q_1 + z$. Form the degree-eight polynomial

$$Q_M(\mu, z) = 4M^5(q_1 + z)^2 \left[H_M(\mu_1 + z/M, \dots, \mu_{M-1} + z/M, z/M) - H_{M-1}(\mu)\right]. \tag{7}$$

It is divisible by z. Because it is symmetric in the μ variables, the coefficient of $z^k \mu^\alpha$ depends only on the partition α of $8 - k$ consisting of its positive exponents. There are

$$p(7) + p(6) + \cdots + p(0) = 15 + 11 + 7 + 5 + 3 + 2 + 1 + 1 = 45$$

possible types. For $M \ge 8$, every such partition fits within the available $M - 1$ variables.

The exact coefficient certificate in Appendix A establishes that all 45 coefficients are positive for every real $M \ge 8$. In particular,

$$Q_M(\mu, z) \ge 0 \qquad (M \ge 8,\ \mu, z \ge 0).$$

Except at the origin, the multiplier $4M^5(q_1 + z)^2$ is positive. Therefore

$$H_M(\mu + z/M, z/M) \ge H_{M-1}(\mu). \tag{8}$$

At the origin both sides vanish.

The cases $1 \le d \le 6$ are already among Bos's proved cases [2, Proposition 3.1]. The accompanying script checks them again exactly, including the one- and two-dimensional bases and degree elevations $8, 5, 3, 3$ for dimensions $3, 4, 5, 6$, respectively. In each finite induction step it checks every monomial type of $(q_1 + z)^r$ times the deficit difference. The multiplier is positive away from the origin; the difference vanishes when $z = 0$. These checks use the same minimum-coordinate reduction as above. Starting with the established case $M = 7$ and applying (8) for $M = 8, 9, \dots$ proves Theorem 1.1.

1.3 What the exact verifier checks

The file `verify_cubic.py` uses only exact SymPy expressions and integers or rational numbers in its assertions. It checks the direct cardinal-polynomial squared-sum identity; the one-dimensional identity

$$H_2(u, v) = 12uv(u^2 - 3uv + v^2)^2;$$

the two-dimensional positive-coefficient base decomposition; all coefficient types in the four remaining low-dimensional induction steps; and all 45 coefficient polynomials in symbolic M after $M = u + 8$.

Its output is:

```
PASS: direct cardinal sum-of-squares identity (six variables).
PASS: base cases, dimensions one and two.
PASS: dimension 3, elevation 8, 123 coefficient types.
PASS: dimension 4, elevation 5, 94 coefficient types.
PASS: dimension 5, elevation 3, 60 coefficient types.
PASS: dimension 6, elevation 3, 64 coefficient types.
PASS: all 45 symbolic coefficient types are nonnegative for M >= 8.
ALL EXACT CHECKS PASSED.
```

A Complete cubic coefficient certificate

For a partition $\alpha \vdash 8-k$, let $C_{k,\alpha}(M)$ be the coefficient of $z^k \mu_1^{\alpha_1} \cdots \mu_r^{\alpha_r}$ in (7). The table gives a positive factor $F_{k,\alpha}(M)$ and an array $(a_0, \ldots, a_s)$ such that the following is an exact polynomial identity:

$$C_{k,\alpha}(8+u) = F_{k,\alpha}(8+u) \sum_{j=0}^{s} a_j u^j. \tag{9}$$

Every displayed a_j is positive, and each displayed $F(M)$ is positive for $M \geq 8$. The empty partition is denoted $\varnothing$; exponents on parts denote repetitions.

k	α	$F_{k,\alpha}(M)$	$(a_0, a_1, \ldots)$
1	(7)	$48M^4\,(M-1)$	(1)
1	$(6,1)$	$120M^4$	$(4,1)$
1	$(5,2)$	$32M^4$	$(108,19)$
1	$(5,1^2)$	$8M^4$	$(691,111)$
1	$(4,3)$	$120M^4$	$(86,13)$
1	$(4,2,1)$	$8M^4$	$(1852,327)$
1	$(4,1^3)$	$24M^4$	$(1681,231)$
1	$(3^2,1)$	$528M^4$	$(37,7)$
1	$(3,2^2)$	$16M^4$	$(246,191)$
1	$(3,2,1^2)$	$16M^4$	$(2933,543)$
1	$(3,1^4)$	$48M^4$	$(3133,438)$
1	$(2^3,1)$	$144M^4$	$(144,59)$
1	$(2^2,1^3)$	$48M^4$	$(2773,493)$
1	$(2,1^5)$	$240M^4$	$(1603,234)$
1	(1^7)	$30240M^4$	$(29,4)$
2	(6)	$36M^3\,(M-1)$	$(22,3)$
2	$(5,1)$	$12M^3$	$(1886,593,43)$
2	$(4,2)$	$4M^3$	$(23766,6679,461)$
2	$(4,1^2)$	$4M^3$	$(46558,13169,879)$
2	(3^2)	$8M^3$	$(19494,5287,359)$
2	$(3,2,1)$	$16M^3$	$(14394,4748,355)$
2	$(3,1^3)$	$24M^3$	$(29968,8563,576)$
2	(2^3)	$24M^3$	$(34,1660,193)$
2	$(2^2,1^2)$	$8M^3$	$(65834,23543,1817)$
2	$(2,1^4)$	$72M^3$	$(24308,7179,498)$
2	(1^6)	$720M^3$	$(5816,1611,108)$
3	(5)	$4M^2\,(M-1)$	$(2538,711,47)$
3	$(4,1)$	$4M^2$	$(117366,49147,6686,295)$
3	$(3,2)$	$8M^2$	$(102258,42029,5739,258)$
3	$(3,1^2)$	$8M^2$	$(234260,99829,13674,605)$
3	$(2^2,1)$	$8M^2$	$(159730,81927,12648,611)$
3	$(2,1^3)$	$24M^2$	$(177980,80093,11338,515)$
3	(1^5)	$480M^2$	$(21806,9274,1263,56)$
4	(4)	$M\,(M-1)$	$(130056,52702,6921,295)$
4	$(3,1)$	$2M$	$(1844808,1015586,206723,18410,605)$
4	(2^2)	$2M$	$(1571928,903466,192249,17862,611)$
4	$(2,1^2)$	$2M$	$(4011400,2362530,501963,45998,1545)$
4	(1^4)	$24M$	$(792080,454696,94306,8465,280)$
5	(3)	$2\,(M-1)$	$(420984,224198,44113,3800,121)$
5	$(2,1)$	6	$(2183112,1536018,425265,57927,3887,103)$
5	(1^3)	12	$(2386440,1698730,470883,63852,4255,112)$
6	(2)	$M-1$	$(325512,179338,36255,3184,103)$
6	(1^2)	2	$(2335608,1664406,463769,63260,4237,112)$

k	α	$F_{k,\alpha}(M)$	$(a_0, a_1, \ldots)$
7	(1)	$8\,(M-3)\,(M-1)$	$(2376, 874, 103, 4)$
8	$\varnothing$	$(M-3)\,(M-1)$	$(2376, 874, 103, 4)$

Here is an exact way of checking the table without expanding in an unbounded number of variables. Treat $q_1, \ldots, q_6, z, M$ as formal indeterminates and substitute (6) into (7). For a product $q_{b_1} \cdots q_{b_s}$, define

$$T((b_1, \ldots, b_s), \alpha) = [\mu^\alpha] \prod_{j=1}^{s} \left(\sum_i \mu_i^{b_j} \right).$$

It is computed by the finite recursion

$$\begin{aligned} T((), \alpha) &= \mathbf{1}_{\alpha=0}, \\ T((b_1, \ldots, b_s), \alpha) &= \sum_{i:\alpha_i \geq b_1} T((b_2, \ldots, b_s), \alpha - b_1 e_i). \end{aligned}$$

This counts all assignments of the power-sum factors to the target monomial's variables, including the correct multiplicities for repeated factors. Apply it to each formal monomial in Q_M. The resulting coefficients are the 45 polynomials represented in the table. The accompanying JSON file also gives the full coefficients of each $C_{k,\alpha}(8+u)$ without factor removal.

The separate file `crosscheck_cubic.py` expands the polynomial directly in a multivariate polynomial ring at $M = 8$. All 3,432 nonzero monomial coefficients agree exactly with the 45-type certificate. This cross-check does not use the power-sum coefficient recursion.

The additional pre-submission checker `verify_review.py` checks every printed table row against the JSON certificate and expands the polynomial directly with symbolic M. To do so, retain seven independent μ coordinates and set the remaining ones to zero. In this restriction,

$$M^j p_j = \sum_{i=1}^{7} (M\mu_i + z)^j + (M-7)z^j.$$

Since Q_M has μ-degree at most seven, every monomial type occurs in this restriction. The resulting 3,432 coefficient polynomials agree exactly with the certificate for symbolic M, providing a second dimension-uniform check without the power-sum recursion.

The all-dimensional argument is a uniform symbolic certificate, not a check at a finite list of large dimensions. The quartic conjectures in the same catalogue entry are not part of Theorem 1.1.

References

[1] M. Colbrook, AIM catalogue, entry 641: Bos's cubic and quartic simplex interpolation nodes. Repository statement accessed 3 October 2026.

[2] L. Bos, *On Fekete points for a real simplex*, Indagationes Mathematicae **34**(2) (2023), 274–293. arXiv:2205.06498v1; doi:10.1016/j.indag.2022.11.003. Section 3, Proposition 3.1, and Appendix A.